IH Used Paper B-TECH-28-10-20

R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, October - 2020
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Give an unambiguous grammar for if-then-else.
b) Explain in detail about various language implementation methods. [7+8]

2.a) Compute the weakest precondition for the following sequences of assignment statements and their post conditions:

```
a = 2 * b + 1;
b = a - 3
{ b < 0 }
```

b) Differentiate between functional and imperative languages. [7+8]

3.a) Discuss different data types in different programming languages.
b) Differentiate between pointers and reference types with examples. [7+8]

4.a) Explain the problems associated with Unconditional Branching.
b) Describe the process of Array initialization. [7+8]

5.a) Describe the problem of passing multi-dimensioned arrays as parameters. Explain with examples.
b) In what fundamental ways do the generic parameters to a Java 5.0 generic method differ from those of C++ methods? [7+8]

6.a) What are the design issues for subprograms? What are the differences between a function and a procedure?
b) What languages allow the user to overload operators? Give examples. [7+8]

7.a) How are instances of Ada generic classes and C++ template classes created? Explain with examples.
b) How can exceptions explicitly raised in C++? Explain with an example. [7+8]

8.a) What is the difference between an ML **val** statement and an assignment statement in C?
b) How exceptions are handled in Python? [7+8]

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